

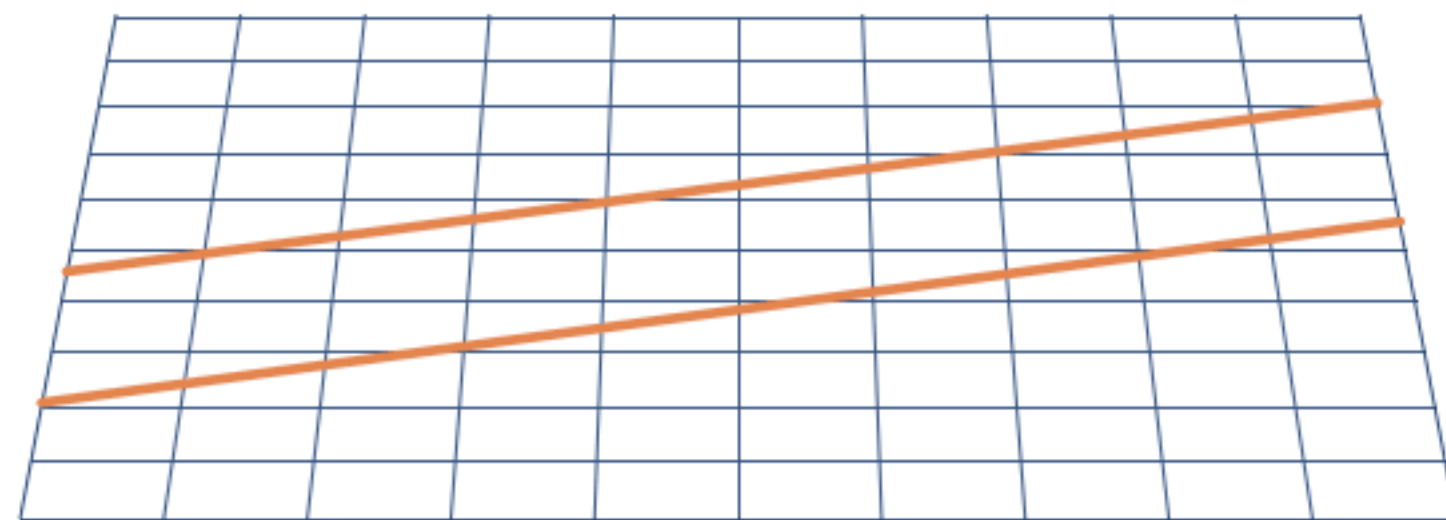
02

The Geometry of Distance

How far apart are two configurations?

On familiar spaces, distance looks the same in every direction
— and the same at every point.

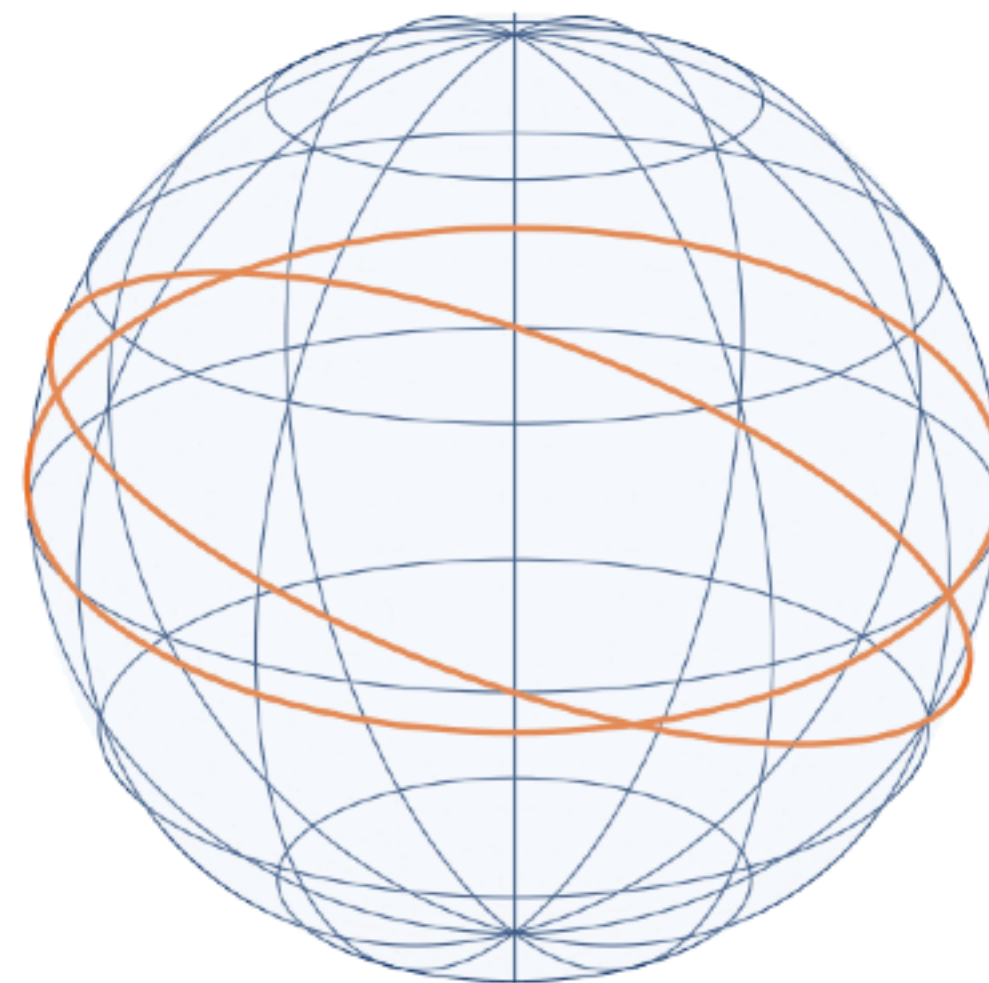
Euclidean plane $\mathbb{R}^2$



$$ds^2 = dx^2 + dy^2$$

geodesics: straight lines

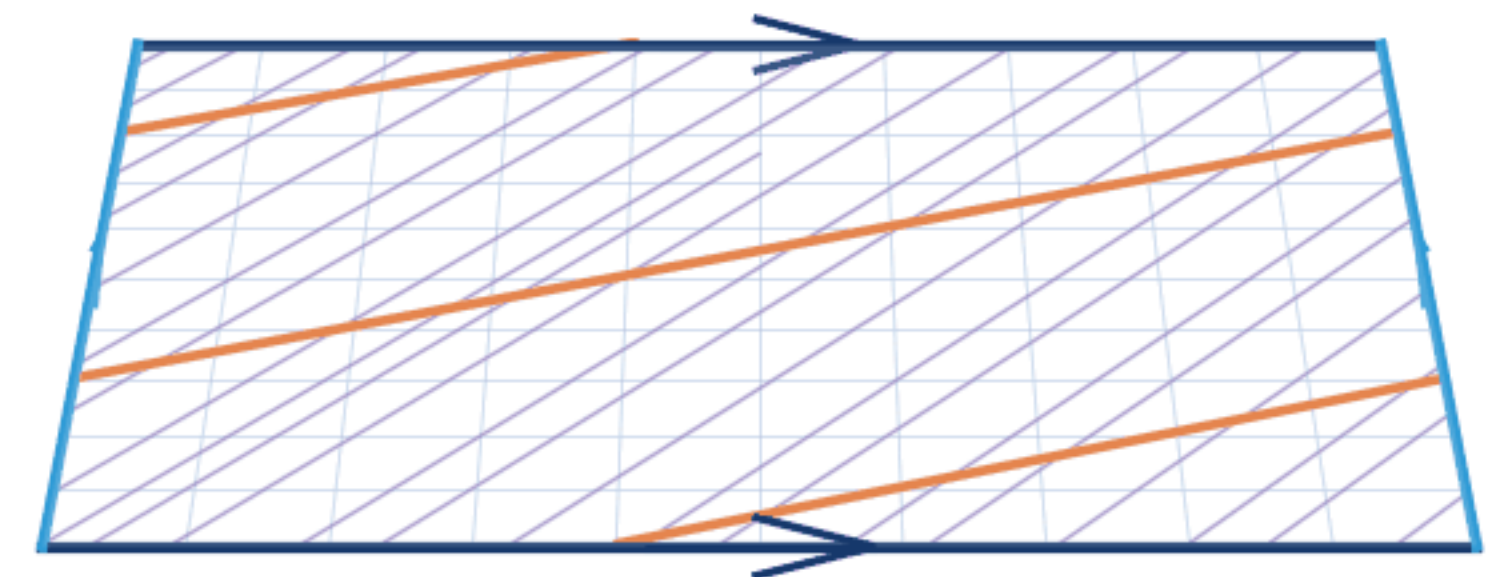
Round sphere S^2



$$ds^2 = d\theta^2 + \sin^2 \theta d\varphi^2$$

geodesics: great circles

Flat torus $\mathbb{R}^2 / \mathbb{Z}^2$



$$ds^2 = dx^2 + dy^2 \quad (\text{locally Euclidean})$$

geodesics: straight lines